

FSOC VIRTUAL CAMERA TRACKING SYSTEM

TARGET: TGT-01

Coarse Alignment - Point-in-Time Performance Report

T = 35.00 s

1. TEST CONFIGURATION

Scenario

-

Target

TGT-01

Camera

640 x 480

FOV

4 deg x 3 deg

Update Rate

30.0 Hz

Noise: None 10.00 | Camera jitter: OFF 12.00 px/frame | Platform motion: OFF 10.00 px | Turbulence: OFF 12.00 px

2. PERFORMANCE

Processing FPS

168.3

>= 20 FPS

Average Tracking Error

3.00 px

Maximum Tracking Error

3.00 px

<= 10 px

Acquisition Time

0.20 s

<= 2 s

Re-acquisition

- s

<= 1 s

Lock Retention

- %

RMSE

- px

Target Loss

0.00 %

Avg Processing

- ms

Re-acquisitions

0

3. PS REQUIREMENT CHECK

Parameter	Measured	Requirement	Status
Update Rate	30.0 Hz	>= 20 Hz	PASS
Processing Speed	168.3 FPS	>= 20 FPS	PASS
Acquisition Time	0.20 s	<= 2 s	PASS
Tracking Error (Max)	3.00 px	<= 10 px	PASS
Target Loss	0.00 %	< 5 %	PASS
Re-acquisition	- s	<= 1 s	-

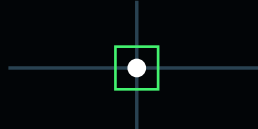
4. VISUAL SNAPSHOT (T = 35.00 s)

RADAR



Target / Boresight / Prediction

VIRTUAL CAMERA



Target TGT-01 • Error 3.00 px

3D DIGITAL TWIN



Camera terminal -> designated beacon

State: TRACKING | Pan: -0.68 deg | Tilt: 0.31 deg | AI: 0.79

5. SYSTEM SUMMARY

HANDOFF READY - COARSE ALIGNMENT GATE SATISFIED

Designated target TGT-01 at 35.00 s. Detection, AI verification, Kalman prediction and pan/tilt control are reported from the recorded telemetry snapshot.